



COM3021/COMM115

Data Science at Scale

Module Overview

DR HUGO BARBOSA

About me

- Dr Hugo Barbosa

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Urban Systems*

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Module Leader COM3021/COMM115

Module Overview

- Module codes: COM3021/COMM115
- ELE Page: <https://ele.exeter.ac.uk/course/view.php?id=18776>
- Contact hours:

Lectures

Tuesdays - 10:35 – 11:25

Lovelace – IC2

Fridays - 15:35 – 16:25

Lovelace – IC2

Workshops

Fridays - 12:35-13:25

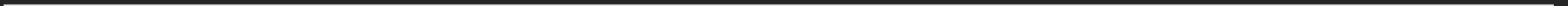
Lovelace – IC2

9 workshop sessions

No workshop on weeks 1 and 12

Module Overview

- Office hours:
 - *Request-based*
 - Send me a message on teams



Syllabus plan

- Introduction: the size of data and impediments to efficient computation;
 - Data storage and retrieval: relational databases and NoSQL systems;
 - Distributed systems and data
 - The MapReduce paradigm and implementations;
 - Algorithms for large scale learning: stochastic gradient descent, large scale linear algebra;
 - Stream processing;
 - Future architectures; co-design of hardware and algorithm
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Assessment

	Weight	Set	Deadline
Coursework	30%	9th Oct	12pm 6 th Nov
Exam	70%	May 2026	

Contacts

- Lectures

Concepts, Theories, Models, Methods

- Workshops

Technologies, frameworks, solutions

Google Cloud Platform

Spark/TensorFlow
